



Volunteer manual

for the healthy lifestyle:
noncommunicable diseases (NCDs)
prevention and control module



International Federation
of Red Cross and Red Crescent Societies
www.ifrc.org Saving lives, changing minds.

In partnership with:



International Federation
of Pharmaceutical
Manufacturers & Associations

**© International Federation of Red Cross
and Red Crescent Societies, Geneva, 2014**

Copies of all or part of this study may be made for noncommercial use, providing the source is acknowledged. The IFRC would appreciate receiving details of its use. Requests for commercial reproduction should be directed to the IFRC at secretariat@ifrc.org.

The opinions and recommendations expressed in this study do not necessarily represent the official policy of the IFRC or of individual National Red Cross or Red Crescent Societies. The designations and maps used do not imply the expression of any opinion on the part of the International Federation or National Societies concerning the legal status of a territory or of its authorities. All photos used in this study are copyright of the IFRC unless otherwise indicated. Cover photo: Ombretta Baggio/IFRC.

P.O. Box 303
CH-1211 Geneva 19
Switzerland
Telephone: +41 22 730 4222
Telefax: +41 22 733 0395
E-mail: secretariat@ifrc.org
Web site: www.ifrc.org

Follow us on:



Healthy lifestyle: noncommunicable diseases (NCDs) prevention and control

module 8 of the CBHFA
manual

The International Federation of Red Cross and Red Crescent Societies (IFRC) is the world's largest volunteer-based humanitarian network, reaching 150 million people each year through our 189 member National Societies. Together, we act before, during and after disasters and health emergencies to meet the needs and improve the lives of vulnerable people. We do so with impartiality as to nationality, race, gender, religious beliefs, class and political opinions.

Guided by *Strategy 2020* – our collective plan of action to tackle the major humanitarian and development

challenges of this decade – we are committed to "saving lives and changing minds".

Our strength lies in our volunteer network, our community-based expertise and our independence and neutrality. We work to improve humanitarian standards, as partners in development and in response to disasters. We persuade decision-makers to act at all times in the interests of vulnerable people. The result: we enable healthy and safe communities, reduce vulnerabilities, strengthen resilience and foster a culture of peace around the world.

Acknowledgments

Special thanks to the following National Societies that contributed to the development of the healthy lifestyle module:

American Red Cross, Armenian Red Cross, Australian Red Cross, Belarus Red Cross, Cambodian Red Cross, Canadian Red Cross, Cook Islands Red Cross Society, Danish Red Cross, Egyptian Red Crescent Society, Finnish Red Cross, French Red Cross, Ghana Red Cross Society, Guatemalan Red Cross, Haitian Red Cross, Indonesian Red Cross Society, Irish Red Cross, Malawi Red Cross, Mongolian Red Cross Society, Norwegian Red Cross, Philippine Red Cross, Qatar Red Crescent Society, Sri Lanka Red Cross Society, Uganda Red Cross, and Venezuelan Red Cross.

Thanks also to the following individuals and Red Cross Red Crescent volunteers and staff:

Project coordination: Ayham Alomari, MD
Design and production: Inart Design Works
Graphic design: Inart Design Works
Training and instructional design/Consultant: Nancy Claxton, Ed.D.

IFRC's global health team
 IFRC's zone health team
 IFRC's Reference Centre for Psychosocial Support team

- 1 http://www.who.int/ncd/mip2000/documents/wha_53_17_en.pdf
- 2 http://www.who.int/nmh/publications/wha_resolution53_14/en/index.html
- 3 http://www.who.int/nmh/publications/wha_resolution56_1/en/index.html
- 4 http://www.who.int/dietphysicalactivity/strategy/eb11344/strategy_english_web.pdf
- 5 <http://www.who.int/nmh/publications/9789241597418/en/>
- 6 http://www.who.int/nmh/publications/un_resolution/en/index.html
- 7 http://www.who.int/substance_abuse/activities/grsua/en/
- 8 http://apps.who.int/gb/ebwha/pdf_files/WHA66/A66_8-en.pdf
- 9 <https://www.un.org/en/ga/ncdmeeting2011/>
- 10 http://apps.who.int/iris/bitstream/10665/94384/1/9789241506236_eng.pdf?ua=1
- 11 http://www.who.int/nmh/global_monitoring_framework/en/
- 12 http://whqlibdoc.who.int/publications/2011/97892406864508_eng.pdf?ua=1
- 13 http://whqlibdoc.who.int/publications/2011/9789241502283_eng.pdf?ua=1

A special thanks is extended to our partner, International Federation of Pharmaceuticals Manufacturers and Associations (IFPMA), which supported the costs for illustration, translation to the four languages and layout.

Finally, we appreciate all the good work previously accomplished, in particular the resolutions on noncommunicable diseases (NCDs) that have been passed by the World Health Organisation (WHO) and the United Nations (UN). The following documents were instrumental in the development of this manual:

- WHO. Resolution (WHA 53.17) adopted by the 53rd World Health Assembly, Prevention and control of noncommunicable diseases, May 2000¹.
- WHO. 53rd World Health Assembly, Report on Global strategy for the prevention and control of noncommunicable diseases, March 2000².
- WHO. Framework Convention on Tobacco Control Resolution (WHA 56.1) adopted by the 56th World Health Assembly, May 2003³.
- WHO. Global Strategy on diet, physical activity and health endorsed by 57th World Health Assembly, May 2004⁴.
- WHO. 2008-2013 Action plan for the global strategy for the prevention and control of noncommunicable diseases, 2009⁵.
- WHO. Resolution (WHA 64/265) adopted by the 64th World Health Assembly, Prevention and control of noncommunicable diseases, May 2010⁶.
- WHO. Global strategy to reduce harmful use of alcohol, 2010⁷.
- WHO. Draft comprehensive global monitoring framework and targets for the prevention and control of noncommunicable diseases, March 2013⁸.
- Political Declaration of the High-level meeting of the UN General Assembly on the Prevention and Control of Noncommunicable Diseases 2011⁹.
- WHO Global Action Plan for the prevention and control of noncommunicable diseases 2013-2020¹⁰.
- NCD Global Monitoring Framework 2025¹¹.
- WHO global status report on noncommunicable diseases 2010¹².
- Noncommunicable disease country profile 2011¹³.

Table of contents

Foreword **5**

Introduction **7**

How to use this manual	7
Learning objectives	9

Topic 1 - The four NCDs: cardiovascular diseases, diabetes, cancers and chronic respiratory diseases **11**

Learning objectives	11
Main learning points	11
The four NCDs	11
Topic summary	12
Cardiovascular disease	13
Cancer	13
Chronic respiratory disease	14
Diabetes	15
Checking for understanding	16

Topic 2 - Identification of four risk factors of NCDs **17**

Learning objectives	17
Main learning points	17
The four risk factors	17
Topic summary	18
Tobacco use	19
Excess alcohol	20
Unhealthy diet	20
Physical inactivity	21
Healthy lifestyle game	22
Checking for understanding	22

Topic 3 - Assessment, identification and facilitation of personal NCD risks **24**

Learning objectives	24
Main learning points	24
Topic summary	25
Risk factor assessment	25
Self-assessment	26
Assisted assessment	27
Clinical assessment	29
Teaching a community group to use the assessment cards	29
Checking for understanding	30

Topic 4 - Identification of risky health behaviours and adoption of healthier behaviours	31
Learning objectives	31
Main learning points	31
Topic summary	32
Behaviour change	32
Slides and ladders of behaviour change	36
Behaviour change stages	36
Checking your understanding	36
.....	
Topic 5 - Development of an health action plan	38
Learning objectives	38
Main learning points	38
Topic summary	39
Health action plans	39
Writing a health action plan	40
Healing your body through behaviour change	45
Checking your understanding	48
.....	

.....

Foreword

.....

The International Federation of Red Cross and Red Crescent Societies (IFRC) and its member National Societies have the unique ability to effectively address communities locally - where they work and live - through a network of more than 15 million volunteers in 189 countries. These community-based volunteers are the true heroes in bringing aid and assistance to their fellow community members, in times of disaster and strife as in peacetime, and are the lifeblood of the Red Cross Red Crescent's important work globally.

The combined threat of infectious diseases, noncommunicable diseases (NCDs) and violence is increasing, and is damaging the health and livelihood of people and communities worldwide. There is an urgent need to scale-up public health interventions and to link them with a broader range of resilience-building initiatives and development areas in order to meet the increasing demand for disease prevention and control.

Being non-infectious by definition, NCDs are largely caused by genetic or lifestyle factors. Four types of NCDs –cardiovascular diseases, cancers, diabetes and chronic respiratory diseases – account for almost two-thirds of the total number of deaths worldwide, with 80 per cent of these deaths occurring in low- and middle-income countries.

NCDs are a leading threat to health and development in all our communities. However, these diseases are preventable.

National Red Cross Red Crescent Societies have a long history of working within communities, and are in a strong position to lead community health initiatives empowering people to take charge of their own health and using tools that have been adapted to their respective local contexts.

There are 98 National Societies currently using the community-based health and first aid (CBHFA) approach to implement community health programming. This programming aims to build the capacity of National Societies, strengthen community systems and improve the health-related behaviour of targeted communities by increasing access to knowledge.

.....

This module is an addition to the existing CBHFA manual, providing materials and guidance to volunteers and community members to help eliminate or reduce these risk factors from community members' lifestyles in a bid to save lives. In doing so, we can help prevent almost 80 percent of heart disease, stroke and type 2 diabetes and over a third of all cancers. It is a task that National Red Cross Red Crescent Societies are uniquely positioned to address. It is the IFRC's goal that this module will help Red Cross Red Crescent volunteers and other partners to deliver these key messages and help promote healthier lifestyles for their communities.



Dr Stefan Seebacher,
Head Health Department
International Federation of Red Cross and Red Crescent Societies (IFRC)

Introduction

How to use this manual

The volunteer manual for the healthy lifestyle: noncommunicable diseases (NCDs) prevention and control module is designed to be facilitated by staff or volunteers. The audience is intended to be community members in any setting.

Relevant information from the volunteer manual is reproduced on the back of the respective visual tools provided in the community toolkit. This enables the volunteers to teach if they may need help with the script for what to say. The security of having all the information right at their fingertips can help in boosting the confidence level of volunteers enabling them to conduct error-free sessions. Volunteers should adapt the materials to best suit the local context but should ensure that modifications keep with the evidence-based messages.

The volunteer manual is intended to be used in conjunction with the community toolkit.

Time required

The volunteer manual outlines the topics for five different community sessions. Teaching the material over five different sessions allows for volunteers and staff to teach the dangers of NCDs, the behaviours that contribute to NCDs, the basics of behaviour change, assessing their NCD risk and developing a health action plan. The tools that are used are highly interactive and appropriate to all audiences.

Topic 1: The four NCDs: cardiovascular diseases, diabetes, cancers and chronic respiratory diseases

Section	Suggested time
Introduction	5 minutes
The four NCDs	20 minutes
Knowledge about NCDs	5 minutes
NCDs puzzle	25 minutes
Total Time	55 minutes

Topic 2: Identification of four risk factors of NCDs

Section	Suggested time
Introduction	5 minutes
The four risk factors	20 minutes
The Dangerous Four (round 1)	20 minutes
The Dangerous Four (round 2)	20 minutes
Healthy lifestyle game	30 minutes
Wrap-up	5 minutes
Total Time	1 hour, 40 minutes

Topic 3: Assessment, identification and facilitation of personal NCD risks

Section	Suggested time
Introduction	5 minutes
Risk factor assessment	20 minutes

Self-assessment demonstration	15 minutes
Assisted assessment demonstration	15 minutes
Clinical assessment demonstration	15 minutes
Wrap-up	5 minutes
Total Time	1 hour, 15 minutes

Topic 4: Identification of risky health behaviours and adoption of healthier behaviours

Section	Suggested time
Introduction	5 minutes
Behaviour change	20 minutes
Slides and steps of behaviour change	15 minutes
Behaviour change stages game	30 minutes
Wrap-up	5 minutes
Total Time	1 hour, 15 minutes

Topic 5: Development of an health action plan

Section	Suggested time
Introduction	5 minutes
Health action plans	20 minutes
Writing a health action plan	30 minutes
Healing your body through behaviour change	30 minutes
Wrap-up	5 minutes
Total Time	1 hour, 30 minutes

Learning objectives

By the end of the module, all beneficiaries will be able to:

Topic 1

- Identify the four principal NCDs of cardiovascular diseases, diabetes, cancers and chronic respiratory diseases.
- Explain the threat of the four NCDs to personal health.

Topic 2

- Identify the four main behavioural risk factors of tobacco use, physical inactivity, harmful use of alcohol and unhealthy diet.
- Share and discuss the specific dangers of tobacco use at the personal and community level.
- Share and discuss the specific dangers of physical inactivity.
- Share and discuss the specific dangers of harmful use of alcohol at the personal and community level.
- Share and discuss the specific dangers of an unhealthy diet.
- Share the economic impact of NCDs at the household level.
- Discuss specific prevention measures of NCDs.

Topic 3

- Identify and discuss factors that correlate to low-, medium- and high-risk levels for NCDs.
- Use the self-assessment scorecards.
- Teach others to use the self-assessment cards.
- Analyse data from the scorecards to determine NCD risk level and next steps.
- Teach others how to analyse data from the scorecards to determine NCD risk level and next steps.

Topic 4

- Identify the volunteer's role as a change agent.
- Discuss the stages of behaviour change.
- Discuss the fluctuations in how people change their behaviours.

Topic 5

- State the three essential elements of effective behaviour change.
- Use scorecard results to determine behaviour change goals and health action plan
- Write a personal health action plan.
- Discuss the positive effects of changing risk factor behaviours.
- Discuss how to best maintain healthy behaviour.

Topic 1

The four NCDs: cardiovascular diseases, diabetes, cancers and chronic respiratory diseases

Noncommunicable diseases are known as NCDs. The four main NCDs are cardiovascular diseases, diabetes, cancers and chronic respiratory diseases.

Learning objectives

At the completion of this topic, you will be able to:

- Identify the four principal NCDs of cardiovascular disease (including heart attacks and strokes), cancers, chronic respiratory disease (like asthma and chronic obstructive pulmonary disease) and diabetes
- Explain the threat of the four NCDs to personal health

Main learning points

Noncommunicable diseases

There has been a strong move in recent years toward inter-agency contingency planning and it is important that Red Cross and Red Crescent National Societies participate in such processes.³ National Societies might be requested to lead the shelter component of inter-agency contingency planning exercises on behalf of the IFRC.⁴ Having internal plans in place will help National Societies to be clear about what they can offer. They should also be prepared to revise internal plans as a consequence of inter-agency planning.

- Noncommunicable diseases (NCDs) are diseases that can NOT be spread through contact with other people or animals that are sick.
- NCDs usually occur when a person consistently practices poor health behaviours like tobacco use, harmful use of alcohol, eating too many unhealthy foods and/or being physically inactive.

The four NCDs

- The four key NCDs are cardiovascular diseases, cancers, chronic respiratory diseases and diabetes.
- NCDs are the number one killer in the world, with more than 63 per cent of all global deaths occurring due to cardiovascular and chronic lung diseases, cancers and diabetes.¹
- You should be prepared to address any medical emergency caused due to an NCD, including providing immediate first aid until they can be seen by a

¹ WHO. Key messages-Noncommunicable diseases (NCDs). World Health Organization, Regional office for Africa, 2012. Available online: <http://www.afro.who.int/en/clusters-a-programmes/dpc/non-communicable-diseases-managementndm/npc-features/3222-key-messages.html>

doctor or clinic. NCD emergencies include strokes, heart attacks and diabetic emergencies.

Cardiovascular disease

- Cardiovascular diseases are a group of problems that occur when the heart and blood vessels work too hard to get blood through your heart and vessels, bringing oxygen to your body.
- This hard work by your heart leads to high blood pressure. These diseases include heart attack and stroke.
- Cardiovascular diseases are mainly caused by a blockage that prevents blood from flowing to the heart or brain. This blockage is often caused by tobacco use, harmful use of alcohol, eating too many unhealthy foods and/or being inactive.

Cancer

- Cancer occurs when a single cell in one part of your body starts to grow abnormally and then grows uncontrolled in your body.
- Tobacco use, harmful use of alcohol, eating too many unhealthy foods and/or being inactive can lead to many kinds of cancers.

Chronic respiratory diseases

- Chronic respiratory diseases occur when your airway and/or lungs become sick and make breathing difficult.
- Chronic respiratory diseases can occur when you use tobacco, drink too much alcohol, eat too many unhealthy foods and/or inhale too much smoke when another person is smoking nearby or even the smoke from burning wood or from charcoal-burning stoves.

Diabetes

- Diabetes occurs when your body cannot control the level of sugar in your blood.
- There are different types of diabetes, but the most common type is called type 2 diabetes. People who have excess body weight, eat too much sugar and/or are physically inactive are at higher risk for type 2 diabetes.

Topic summary

Knowledge about NCDs

Tool 1.1 - NCDs diagram poster

The four major NCDs that globally cause the most number of deaths each year are cardiovascular diseases, diabetes, cancer and chronic respiratory diseases. Most NCDs are preventable when people:

- Increase healthy behaviours, including:
 - being active on a daily basis
 - eating healthy foods
- Reduce or eliminate unhealthy behaviours, including:
 - Using tobacco products
 - Drinking too much alcohol

Tool 1.2 - NCDs puzzle

- Blank puzzle board
- NCD and risk factor puzzle pieces (8 in total)

Cardiovascular disease

What is it?

- Cardiovascular disease is often caused by excessive eating of unhealthy foods, using tobacco, drinking too much alcohol and/or not doing enough physical activity.
- Heart attacks happen when a part of the heart muscle dies or gets damaged because there is not enough blood reaching the heart. Blood vessels cannot provide enough blood to the heart when the arteries or blood vessels are blocked.²
- Strokes occur when a blood vessel in the brain becomes blocked or bleeds.

What happens if I have cardiovascular disease?

- If you eat a lot of unhealthy foods, your blood vessels become blocked. This makes your heart have to work too hard.
- Physical activity will help you to clear the blockages in your blood vessels.
- If you have a cardiovascular disease, you should talk to your doctor and change your health behaviours soon.

How can I prevent or treat cardiovascular disease?

- Being physically active helps to clear some of the blockages in the vessels and heart.
- Get your blood pressure measured regularly. The blood pressure tells you how much blockage you have in your arteries by measuring how hard your heart has to work to pump blood.
- Know that cardiovascular disease can be prevented and to avoid heart attack and stroke of the brain:
 - a. Stop using tobacco
 - b. Eat five servings of fruits and vegetables every day
 - c. Reduce the amount of fat and salt in your diet
 - d. Get at least 30 minutes of physical activity per day
 - e. Do not drink excessive amounts of alcohol
- Stress and worry can also increase your risk for cardiovascular disease. Meditate, exercise and enjoy time with friends and family to help reduce your stress.
- If you have cardiovascular disease, take the medicines based on your doctor's advice. Get regular check-ups done.

Cancer

What is it?

- Cancer occurs when a single cell in any part of the body starts to grow abnormally and then grows uncontrollably.
- Cancer starts to grow in one organ and then can spread to other organs. When cancer grows in an organ, that organ cannot function normally and can even stop.
- Cancer is a deadly disease that can affect people from any race, gender, class or nationality.
- At least one-third of all cancer cases diagnosed are preventable. These pre-

² CDC. Heart disease. Centers for diseases control and prevention, 2012. Available online: http://www.cdc.gov/heartdisease/heart_attack.htm

ventable cancer deaths are linked to tobacco use, harmful use of alcohol, excessively eating unhealthy foods and being inactive.³

- Cancer can occur in different organs of the body including blood and skin. Lung cancer is one of the deadliest cancers and is most common in smokers and people who live and work with smokers.

What happens if I have cancer?

- When cancer starts to grow in your body, cancer cells grow very quickly. If untreated, the cancer will quickly spread and kill.
- If you have been diagnosed with cancer, you must seek medical attention immediately.

How can I prevent or treat cancer?

- Out of all cancer cases, 33 per cent of cases can be prevented by stopping the use of tobacco, alcohol and unhealthy foods, increasing physical activity and eating more fruits and vegetables.
- Early diagnosis helps improve the likelihood of surviving cancer.
 - o Women should get regular check-ups.
 - **To screen for breast cancer:**
 - Women more than 20 years of age should practice monthly breast self-exams. Self-exams can help to detect over 45 per cent of breast tumours.⁴
 - Women aged between 50 and 74 should get a mammogram in every 2 years. If there is a higher risk because of family history or past cancer diagnoses, your doctor will advise if you should schedule a mammogram before 50 years of age.
 - **To screen for cervical cancer:**
 - Women more than age of 21 years should get a regular Pap test.
 - o Men should get regular check-ups and screening for cancers, as appropriate.
- Treatments include:
 - a. Surgery to remove the cancer cells or a complete organ that is diseased.
 - b. Undergoing chemotherapy and radiotherapy.

Chronic respiratory disease

What is it?

- Chronic respiratory diseases occur when your airway and/or lungs become sick and make breathing difficult.
- They are usually caused by breathing in tobacco or smoke. The good air that your body needs cannot get to your lungs and organs.
- Chronic respiratory diseases can also occur when you inhale smoke from someone else's cigar or cigarette.
- You can also get a chronic respiratory disease when you inhale smoke regularly from a fire burning wood, animal dung or crop waste.

What happens if I have chronic respiratory disease?

- When not treated, it is very difficult to breathe and you will often have a feeling that you cannot get enough air to survive. It is a very scary feeling.
- When treated, medicines can help your lungs get the air you need for short periods of time.
- Repeated smoking or breathing polluted air will dampen the chances of your lungs to recover and heal.

³ WHO, *Cancer Prevention*, 2013.

⁴ UICC. *Simply check campaign for detecting breast cancer*, Union for International Cancer Control, 2012. Available online: <http://www.uicc.org/simply-check-campaign-detecting-breast-cancer>

- A person who has a chronic respiratory disease can become worse if s/he uses tobacco, drinks harmful amounts of alcohol, excessively eats unhealthy foods and/or is physically inactive.

How can I prevent or treat chronic respiratory disease?

- Completely stop using tobacco.
- Stop breathing second-hand smoke from others who are smoking.
- Stop breathing smoke from fires fuelled by wood or animal dung, or crop waste.⁵
- Children and youth are especially vulnerable to second-hand smoke because their bodies are still developing.
- While physical activity may be difficult, mild activity will help to prevent or ease even further problems associated with chronic respiratory diseases.
- By eating a healthy diet with plenty of fruits and vegetables, being physically active, reducing alcohol intake and eliminating tobacco and smoke inhalation, a person's body can be made stronger against complications from chronic respiratory diseases.

Diabetes

What is it?

- Diabetes occurs when your body cannot control the level of sugar in your blood.
- There are different types of diabetes, but the most common type – called type 2 – affects people who have excess body weight, smoke tobacco, drink harmful amounts of alcohol, eat an unhealthy diet and/or are physically inactive.
- Diabetes can affect people of any age, race, gender or nationality.
- Adults need to monitor their own sugar, fats and calorie intake and physical activity levels and monitor the sugar intake and activity levels of their children too.

What happens if I have diabetes?

- When you have diabetes, you may feel like you need to urinate often, you may be extremely thirsty or hungry all the time, feel extremely tired, have cuts or bruises that take a long time to heal or have a tingly feeling in your hands and feet.
- Sometimes, people can have diabetes and not even know they have it. Getting your blood sugar tested is the best way to see if you are at risk.
- Having diabetes can double your risk of heart attacks and strokes of the brain, failure of the kidneys and even blockages in your blood vessels and nervous system that can mean amputation of the feet and/or loss of eyesight.

How can I prevent or treat diabetes?

- Treatment for people with diabetes includes reducing the amount of sugar, fat and calories they eat and drink.
- Diabetics usually also have to give themselves pills or injections every day.
- Diabetes medication is expensive and puts many families in debt. It requires regular purchases of needles, testing kits and bottles of insulin for the rest of your life or as long as you have the disease.
- Type 2 diabetes, caused by an unhealthy lifestyle, can be prevented through adopting and maintaining healthy behaviours.
- If you already have type 2 diabetes, you can reverse the effects of the disease by adopting and maintaining healthy behaviours.
- To stop the effects of type 2 diabetes, you would need to stop eating sugary, fat

⁵ WHO. Fact sheet-Indoor air pollution and health, 2011. Available online: <http://www.who.int/mediacentre/factsheets/fs292/en/>

and high-calorie foods and drinks and become more active.

- You also need to stop the use of tobacco products and drinking harmful amounts of alcohol.

Tools 1.3, 1.4 and 1.5 - First aid in NCD emergencies

- **First aid for a person suffering a cardiac arrest poster**
- **First aid for a person suffering a stroke poster**
- **First aid for a person suffering diabetic emergency poster**

People, who have an NCD, may not always know that they have a disease. There are different emergencies that people with an NCD may experience such as a heart attack, a stroke or a diabetic emergency. Share the posters and the symptoms and treatments for three such NCD emergencies.

Checking for understanding

1. What are the four main NCDs that are responsible for 63 per cent of global deaths every year?
2. What are the four risk factors for contracting these four NCDs?
3. Do any of your family members practice any of these unhealthy behaviours?

Case study:



The Qatar Red Crescent Society recognized that many of Qatar's residents suffer from respiratory diseases as a result of the constant construction occurring in Qatar's growing city. The workers helping to build the new buildings in Qatar are largely foreign workers; roughly 15 per cent of these workers are at high risk for NCDs or have been already diagnosed with an NCD. The unhealthy habits that are the biggest problem in Qatar are tobacco use, unhealthy diet and poor quality of air in addition to a general lack of information about how these risks affect their health. In cooperation with the Ministries of Health and Labour, the Qatar Red Crescent Society opened the Workers Health Centre in 2010 to work with communities and individuals to identify their risk factors and minimize the number of NCDs through health promotion, education and behavioural changes.

Topic 2

Identification of four risk factors of NCDs

Learning objectives

At the completion of this topic, you will be able to:

- Identify the four main behavioural risk factors of tobacco use, excessive use of alcohol, unhealthy diet and physical inactivity
- Share and discuss the specific dangers of tobacco use at the personal, household and community level
- Share and discuss the specific dangers of physical inactivity
- Share and discuss the specific dangers of excessive use of alcohol at the personal, household and community level
- Share and discuss the specific dangers of an unhealthy diet
- Share the economic impact of NCDs at the household level
- Discuss specific prevention measures for NCDs

Main learning points

- There are four main behaviours that predispose the body to being diagnosed with an NCD. These behaviours include tobacco use, physical inactivity, harmful use of alcohol and unhealthy diet.
- Especially, parents can positively influence their children's health through avoiding these behaviours and teaching healthy behaviours.

The four risk factors

- **Tobacco** is globally the single greatest cause of preventable death. Tobacco use affects an entire community.⁶
- Tobacco can be found in many forms such as cigarettes, cigars, bidis, pipes, kreteks, chewing tobacco and shishas or water pipes.⁷
- Continually using tobacco and inhaling smoke contributes to your risk for NCDs.
- Tobacco use affects an entire community – breathing in the smoke from someone else's tobacco is even more dangerous than it is for the smoker.
- Tobacco smoke contains dangerous chemicals that cause many genetic mutations in your body. Tobacco also contains nicotine, which is highly addictive.⁸ In women, tobacco smoke can cause miscarriages.⁹ In men, tobacco use can greatly increase the risk of impotence.¹⁰

⁶ WHO. *Report on the Global Tobacco Epidemic, 2008: The MPOWER package*. Geneva, World Health Organization, 2008. Available online; http://www.who.int/tobacco/mpower/mpower_report_full_2008.pdf

⁷ Eriksen M and Mackay J. *The Tobacco Atlas*. World Health Organization, 2002. Available online; <http://whqlibdoc.who.int/publications/2002/9241562099.pdf>

⁸ WHO. *Gender, women and the tobacco epidemic*. Geneva, World Health Organization, 2010. Available online; http://www.who.int/tobacco/publications/gender/en_tfi_gender_women_addiction_nicotine.pdf

⁹ Farioli, et al., 2010.

¹⁰ Tengs, et al., 2001.

- **Excessive alcohol** intake can disrupt normal brain development and damage organs like liver, brain, stomach, intestines and heart. Too much alcohol also damages that part of the brain that controls coordination, memory, judgement and decision-making. Continually drinking too much alcohol contributes to your risk for NCDs.
 - Excessive alcohol intake affects an entire community. People who drink too much can behave dangerously when they drive, swim, act violently or act in a way that upsets others.
 - In 2004, statistics show that out of all the alcohol-attributable deaths about 20 per cent came from cancer – including 19 per cent males and 25 per cent females; and about 22 per cent came from cardiovascular diseases – including 23 per cent males and 18 per cent females.¹¹
 - By reducing or eliminating alcohol intake, a person can greatly reduce their risk for NCDs. While official guidelines vary by country, restricting drinking to no more than 1 to 2 alcoholic drinks per day is advisable.
-
- An **unhealthy diet** leaves you feeling sick, your body poorly nourished and less able to fight off sickness. By eating a healthy diet, including five servings of fruits and vegetables every day, your body is less likely to become sick as it has the nutrients it needs to make your body strong. Eating a healthy diet also means that you are less likely to eat unhealthy foods that can make you sick, overweight and tired, thereby greatly reducing the risk for NCDs.
 - Reduce the use of salt, sugar or fat in your foods. These can usually be found in most condiments used to flavour foods like seasoning salt, ketchup, dressings, dips, sauces or cream.
-
- **Physical inactivity** causes the body to start to break down. By staying physically active for at least 30 minutes every day, a person can greatly reduce their risk for NCDs.
 - By doing at least 30 minutes of activity every day such as walking or biking, you keep your body healthy and more ready to fight disease. Doing activity every day lowers your risk for NCDs.

Topic summary

Introduction

The four risk factors including tobacco use, harmful amounts of alcohol, unhealthy diet and physical inactivity all contribute to the risk for being diagnosed with an NCD. These four risk factors are common causes for NCDs.

Tool 2.1 - The Dangerous Four posters

- Tobacco use
- Excess alcohol
- Unhealthy diet
- Physical inactivity

¹¹ Parry, et al., 2013.

Tobacco use

Let us look at how tobacco affects the body.

We can tell tobacco users that using tobacco products is the most dangerous behaviour and puts them – and those around them – at the greatest risk for diabetes, cancer, cardiovascular diseases and chronic respiratory diseases.

- Tobacco products contain tar, which damage your lungs, your throat, mouth and your entire body.
- Tar and carbon monoxide produced when tobacco is lit cause damage to the body.
- Nicotine is an addictive material in tobacco that becomes difficult to quit once a person become addicted.¹²
- Smoke from tobacco and other organic materials contain dangerous chemicals that cause many genetic mutations in the body.¹³
- ALL forms of tobacco are deadly, including:
 - Cigarettes or bidis
 - Cigars, also known as cheroots, stumphen or dhumtis
 - Kreteks
 - Clay pipes, also known as suipa, chillum or hookli
 - Water pipes, also known as shisha or hubbly bubbly
 - Chewing tobacco, also known as plug, loose-leaf, snuff or twist¹⁴
- Breathing in smoke from another person's tobacco smoking or even breathing in the smoke from a fire burning from wood, animal dung or crop waste is even MORE dangerous.¹⁵
- Breathing tobacco smoke is especially dangerous for infants and children since their lungs are in the developmental stage and are highly vulnerable.
- In men, tobacco use can greatly increase the risk of impotence.
- With every bit of tobacco used, the damage becomes greater and with continued tobacco use the risk of NCDs increases many folds.
- Younger bodies are more likely to become damaged by tobacco. The younger you are when you start using tobacco, the more damage occurs to your body since a younger body is still developing and is much more vulnerable to tobacco and smoke.

Let us look at how tobacco affects each part of your body.

- Worst of all, tobacco contains tar and nicotine that is addictive. This means that once you start, it will be even harder to stop.
- Tobacco affects your mouth and teeth first. It makes your teeth turn yellow and gives you bad breath.
- Tobacco use means that your skin is more likely to wrinkle and you are more likely to lose your hair at a younger age.
- Tobacco and smoke also causes cancer of the gum, mouth, throat and voice box, as well as lung cancer.
- Smoking tobacco makes you cough and can lead to bronchitis and emphysema.
- Smoking affects your other organs too and predisposes you to be diagnosed with cancer of the bladder, kidney, pancreas, stomach or sex organs.
- Tobacco makes your blood vessels grow smaller making your heart work harder and can lead to another NCD – cardiovascular disease.

What can you do?

- Stop smoking, using tobacco products and/or breathing smoke from other smokers or even from a cooking stove. Breathe clean, fresh air.

¹² WHO. *Report on the Global Tobacco Epidemic, 2008: The MPOWER package*. Geneva, World Health Organization, 2008. Available online: http://www.who.int/tobacco/mpower/mpower_report_full_2008.pdf

¹³ Pfeifer, et al., 2002.

¹⁴ WHO. Eriksen M and Mackay J. *The Tobacco Atlas*. World Health Organization, 2002. Available online: <http://www.who.int/tobacco/en/atas4.pdf>

¹⁵ WHO. *Fact sheet - Indoor air pollution and health, 2011*. Available online: <http://www.who.int/mediacentre/factsheets/fs292/en/>

- Ask for help for smoking cessation from a health clinic or other professionals.

Excess alcohol

Let us look at how excessive amounts of alcohol affect the body.

- Too much alcohol increases your risk of cardiovascular diseases, cancer, liver diseases and diabetes.
- When you drink, the alcohol is absorbed directly into your bloodstream.
- The alcohol in your blood has to be processed by the liver. The liver can only process one standard serving of alcohol per hour.
- Any extra alcohol after one serving is in your bloodstream where it now has direct access to all your organs as it waits for the liver to be ready to process it. As the alcohol-soaked blood flows through every major organ, it causes damage, making you more vulnerable to NCDs like cancer, cardiovascular diseases, chronic respiratory diseases and diabetes. The major organs that are most greatly affected are:
 - a. Liver: Excessive use of alcohol causes the liver to become sick and filled with scar tissue. Too much alcohol can cause cancer of the liver and other organs.
 - b. Digestive organs: Your entire digestive system including your stomach, small intestines, oesophagus and pancreas are all affected by consistent excess of alcohol. Cancer and chronic respiratory diseases are more likely to develop in these organs if you drink too much.
 - c. Reproductive organs:
 - i. In men, too much alcohol can cause impotence.
 - ii. In women, too much alcohol can make it difficult to become pregnant. If a woman is already pregnant, it is likely that drinking alcohol will hurt the unborn baby and can cause birth defects.¹⁶

What can you do?

Avoid alcohol if you can. If you must drink, drink moderately – no more than 1 to 2 servings each day.

- Pregnant women should not drink any alcohol.

Tool 2.2 - "How much is too much?" alcohol serving size poster

Excessive use of alcohol is more than one standard serving size per hour. See the How much is too much? poster for approximate amounts that equal one drink.

Unhealthy diet

Let us look at how unhealthy foods and drinks affect the body.

- When you eat food, your body will digest and process it into energy according to its requirement and whatever is surplus will be stored as fat in the blood vessels, heart and other organs.
- This extra fat blocks your arteries and makes your heart work harder to push your blood through to the rest of your body. Fat also gets stored in your organs that makes them work extra hard to do their job.
- All foods have calories that fuel the body. A high calorie food is not necessarily

¹⁶ Warren, et al., 2001.

a high-grade or high-quality food.

- Fruits and vegetables have a small number of calories but are a high-grade or high-quality fuel. They contain a lot of water and fibre as well as vitamins and minerals that your body needs to feel and be well.
- Some foods that are high in calories like lard, butter, candy, fried foods are low-quality fuel. They usually just contain things that your body does not really need. They contain too much fat, which blocks up your heart and blood vessels, making your heart work harder than it should.
- Sugar, when eaten or drunk, turns to fat and makes your cells sick.
- When your body continually does not get all the vitamins, minerals and water it needs in the foods you eat, your body struggles to do its work well and you will feel tired, exhausted and unwell.
- If an organ in your body does not get all the ingredients it needs from the foods you eat to do its job well, it simply will not do its job well or even will not work at all and is much more likely to become sick and stop working.
- When your body organs do not work well or stop functioning, you can become very sick or even die.
- Drinking sweetened drinks only adds more stress to your body as it tries to process the sugar and puts you at risk for diabetes and cardiovascular disease.

What to do?

- Drink lots of water – about 4 to 5 large glasses every day. This helps your body to wash out the bad things it is trying to get rid of, helps you to defecate more often and gives your body the fluids it needs to work properly.
- Eat five servings of fruits and vegetables every day.
- Limit fat, sugar and salt in your diet.

Tool 2.3 - Healthy eating guide and Fruit and vegetable serving sizes posters

The healthy eating guide poster in your toolkit compares healthy versus unhealthy foods and shows how healthy foods benefit your body and vice versa. The fruit and vegetable serving sizes poster shows what a serving size is. It is adapted to your local context.

Physical inactivity

Let us look at how not doing physical activity affects the body.

- Physical activity keeps muscles, joints, bones, tendons – all your body parts – moving and working to the fullest potential. When you do not use them, they lose their flexibility or ability to do the things they should be able to do. The less you use your muscles, the more muscle you lose. When you do too little activity, your body starts performing poorly and working abnormally.
- When you consistently do less than 30 minutes of activity each day, your lungs produce less oxygen just as your body demands more oxygen, your heart pushes less blood through it, you have a hard time defecating and your bones and muscles actually start to weaken and even break down.
- In this weakened state, your body is vulnerable – it is not working at 100 per cent and so is much more likely to become sick.
- When you sit too long, the activity in your muscles shuts down, your body stops burning energy efficiently and your body loses much of its ability to break down fat in your body.
- The longer you stay inactive, the more your muscles break down, the less able

your body is to pump blood and breathe efficiently and you are at a higher risk of having diabetes, cardiovascular disease, cancer and chronic respiratory disease.

What to do?

- Become a little more active each day.
- Try to move your body as much as you can throughout the day every day.
- Walk to school, the market or work when you can.
- Stretch your muscles by doing stretching exercises or yoga twice a week
- Give your muscles a workout by doing simple exercises that stress the muscle safely to make them stronger like push-ups, sit-ups and squats.
- Do sports that you enjoy.

Tool 2.4 - Exercise cards

Look at the exercise cards for some exercises you can do now.

Healthy lifestyle game

NCDs can personally cost you most of your wages each year. Approximately 100 million people each year are pushed into poverty when they have to pay directly for health care costs partially caused by poor habits.¹⁷ These health care costs come from lost wages or salary when people are too ill to work, to pay for a doctor or clinic fees, a stay in a hospital, medicine and other costs. If you are not yet paying these costs and engage in one or more of the four risk factors, you will likely one day soon be affected by these costs. Let us look at these costs to help you see how unhealthy habits affect your body and ultimately, your wallet.

Tool 2.5 - Healthy lifestyle game

- If working with a large group, play in groups of 3 to 5 players per group. If conducting a home visit, play with any interested family members in one group.
- Distribute to each group a game board, a die, pawns and a stack of game cards. Each player in the group receives 25 Red Bucks (RB).
- Follow the guidance and player directions included in the game.
- The goal is to be the first to lead a long life by successfully reaching the finish line and holding at least 1 Red Buck.
- After the game ends, lead a discussion on their experience playing the game.
 - What did you like about the game?
 - What was frustrating about the game?
 - Who won by living a long life? How did they win?
 - Who died early in the game? What happened?
 - Who ended up with the most money?
 - Who ended up with the least money?
 - Is buying healthy foods that might be more expensive than unhealthy foods worth the extra cost? Explain your response.
 - What costs are associated with unhealthy habits?

Checking for understanding

- What are the main risk factors?
- How does tobacco affect the body and make you vulnerable to NCDs?
- How does too much alcohol affect the body and make you vulnerable to NCDs?
- How does an unhealthy diet affect the body and make you vulnerable to

¹⁷ Providing for Health, 2010.

NCDs?

- How does physical inactivity affect the body and make you more vulnerable to NCDs?
- What are unhealthy foods in your country or region?
- What are healthy foods in your country or region?
- How can NCDs affect you and your family financially? Physically? Emotionally?

Case study:



The Sri Lanka Red Cross Society developed a long-term strategy to address NCDs in their country. The National Society works in schools through Junior Red Cross, organizes national campaigns against drugs, alcohol and tobacco use and a World Diabetes Day event. Volunteers joined in surveillance activities, visited households, invited high-risk individuals to attend a screening programme at the nearest government health centre to test for health risk and conducted community awareness interventions such as health talks and establishing a gym to encourage community members to be more physically active.

The screenings were provided to 619 individuals and 81 per cent of those screened were identified as having diabetes and 49 per cent had high blood pressure. These high-risk individuals were given counselling on behaviour modifications and treatment to try and reverse the disease and the disease's effects on the body.

At the end of the project there was evidence that many community members had changed their unhealthy behaviours to prevent NCDs. These included:

- An increased number of community members use the gymnasium set up at the National Society's Colombo Branch.
- An increased number of people, specifically working parents, are now willing to serve traditional food instead of junk food.
- An increasing number of adults over the age of 40 are getting regular health check-ups done.
- NCD risk factors are being addressed at the household level thanks to the large female beneficiary representation.

Topic 3

Assessment, identification and facilitation of personal NCD risks

Learning objectives

At the completion of this topic, you will be able to:

- Identify and discuss factors that correlate to low-, medium- and high-risk levels for NCDs
- Use the self-assessment cards
- Teach others to use the self-assessment cards
- Analyse data from the scorecards to determine NCD risk level and recommend next steps
- Teach others how to analyse data from the scorecards to determine NCD risk level and recommend next steps
- Provide home-based support to community members at highest risk and as needed

Main learning points

- People who engage in any of the four risk factors are at risk for being diagnosed with an NCD.
- The longer the risky behaviour is practiced; the intensity of behaviour and the age of the person all affect their likelihood for getting an NCD.
- Measuring a person's risk level for NCDs is done in three stages: self-assessment, assisted assessment and clinical assessment.
- There are three levels to assessing your risk for NCDs.
 - o **Self-assessment** of your NCD risk level simply involves answering questions about your current habits and behaviours honestly.
 - o Having an **assisted assessment** for your NCD risk level involves having a volunteer measure your blood pressure using a special cuff and taking your weight and height.
 - o Having a **clinical assessment** for your NCD risk level involves a visit to the health clinic to give a blood sample for analysis.
- If you are healthy but have one or more risk factors for NCDs, any of the three assessments conducted regularly can help to control the onset of disease. Regular assessments help identify disease and allow for proper treatment and control of the disease through behaviour modifications and possibly, medication.
- If you know that you have an NCD and still practice the risky behaviours that caused the disease, regular assessments and changes in behaviour can prevent further complications.
- Changing behaviour is possible. Understanding your motivation and how to use behaviour change techniques helps to make change possible.

- When a person learns that they have an NCD, the news and what it means can be frightening. Providing psychosocial support or follow-up support can help newly diagnosed community members to ease the worry.

Topic summary

Introduction

NCDs are most likely to be diagnosed in people who engage in risky behaviours like using tobacco, drinking excess alcohol, eating unhealthy food and being physically inactive.

Many people engage in some or all of these behaviours occasionally and will be at a different risk level than other people who engage in these behaviours more frequently.

Regularly assessing your risk level helps to understand your risk for having an NCD. Knowing that you have an NCD and taking steps to monitor your disease through behaviour change will help you feel better and live longer. In some cases, long-term medication may be required to control the disease.

Risk factor assessment

Tool 3.1 - NCD risk assessment card

The assessment card identifies three different levels of assessment: self-assessment, assisted assessment and clinical assessment. The card is two-sided with the self-assessment on the front and the assisted assessment and clinical assessment at the back.

The first section – self-assessment – allows a person to assess their own risk for NCDs by answering simple questions. The card prompts a person about their age and sex to see if they are at a risk for developing NCD based on this basic information.

A person is prompted about their health behaviours around tobacco use, alcohol use, healthy foods they eat and physical activity they do. For each answer they give, points are awarded using a weighted system.

Additionally, a person answers questions about age, sex and if a family member has an NCD. Based on their responses, the card provides each person with their risk level and a basic recommendation.

The second section – assisted assessment – requires a person to work with their volunteer to measure their blood pressure and body weight and height to further assess their risk. These measures help to better determine risk level for NCDs. Based on their responses; the card provides each person with their risk-level-based and basic recommendations.

The third section – clinical assessment – is advised for people whose previous assessment scores show that they are at a high risk. The clinical assessment prompts a person to go to an equipped clinic or lab to provide blood samples

for analysis. The clinical assessment tests can determine the level of sugar and cholesterol in the blood. The clinical assessment is helpful and necessary to diagnose NCDs and other diseases.

Self-assessment

Self-assessment requires a person to read the three answer choices next to each behavioural risk factor.

Healthy diet

The definition of a healthy diet is to eat five servings of fruits and vegetables regularly – preferably every day. A person indicates one of the following:

- Eats five servings of fruits and vegetables regularly.
- Eats some fruits and vegetables, but less than five servings regularly.
- Eats very little or no fruits and vegetables regularly.

Physical activity

The definition of physical activity is to do 30 minutes of activity that keeps your body moving every day like walking, biking or, exercise at the gym. A person states one of the following:

- Physically active at least 30 minutes regularly
- Physically active, but less than 30 minutes regularly
- Not at all physically active

Excess alcohol

The definition of use of alcohol is the amount of alcohol a person drinks in 1 day. A person indicates one of the following:

- Men who drink 0 to 2 alcoholic drinks per day/Women who drink 0 to 1 alcoholic drinks per day.
- Men who drink 3 to 4 alcoholic drinks per day/Women who drink 2 to 3 alcoholic drinks per day.
- Men and women who drink more than 4 drinks per day or who occasionally binge drink. Binge drinking is drinking more than 4 drinks within a short period of time.

Tobacco use

The definition of tobacco use is smoking or using tobacco products. A person indicates one of the following:

- Have never used tobacco OR have stopped using tobacco completely more than 10 years ago OR are not exposed to smoke consistently.¹⁸
- Stopped using tobacco less than 2 years ago OR are exposed to smoke once in a while.
- Use tobacco or is exposed to smoke.

Each risk factor answer choice is marked by:

- A smiling healthy face – low risk,
- An expressionless face – medium risk
- A sick face – high risk.

Additionally, three non-behaviour-related questions provide further information:

¹⁸ WHO, *Fact Sheet about the health benefits of smoking cessation, 2013.*

Sex and age

Men older than 45 years and women older than 55 years are at a greater risk of NCDs. All people who meet these conditions are advised to go for an assisted assessment consisting of a blood pressure check every 2 years, regardless of health behaviour.

Family history of NCDs

If a biological family member – a mother, father, brother, or sister has been diagnosed with an NCD, a person is also at greater risk. If known, this information will help to determine a person's risk level more accurately.

Self-assessment score

A simple scoring indication is provided on the card to allow the person to find their risk level. This self-assessment does not need to be shared with volunteers, but allows a person to see how their behaviours relate to their NCD risk level. Community members can be taught to use the self-assessment cards with little assistance. They should be referred to the appropriate community resources for the assisted and clinical assessment. Research the resources in your area for where people can go for their assisted and clinical assessments.

Look in the toolkit for the suggested process to help people to assess at each level of assessment and next steps.

Assisted assessment

Assisted assessment requires minimal equipment including a tape measure (included), a weight scale and a blood pressure cuff. The tape measure is for measuring a person's height. The scale is for measuring body weight. Both of these measures find a person's body mass index (BMI) that determines the general healthy weight for a person. The blood pressure cuff is for measuring blood pressure.

Waist circumference

Tool 3.2 - Measuring strip

If a person has too much fat on their belly or abdomen, it adds to their risk. By measuring the circumference of the waist correctly, you can determine if you carry too much fat around your belly. Too much fat around your abdomen increases your risk for disease.

Body Mass Index (BMI)

Tool 3.2 and 3.3 - Measuring strip and Body Mass Index (BMI) chart

BMI is a measure of both height and weight to determine if a person 18 years and older is underweight, overweight or at the appropriate weight. To determine BMI, find the adult's body weight and height using a scale and a tape measure. Once you have this information, use the BMI chart in the toolkit to determine the adult's BMI.

Note that there are different scales for measuring the BMI of children that vary by country.¹⁹

The measures shaded in green indicate that the person has a BMI between 18.5 and 25 and has a lower risk level for NCDs based on their body weight. The measures shaded in yellow indicate that the person has a BMI between 25.1 and 30 and has a medium risk level for NCDs based on their body weight. The measures shaded in red indicate that the person has a BMI higher than 30 and has a high risk level for NCDs based on their body weight.

Blood pressure

If your team has access to an automatic blood pressure monitor, you will be able to provide assisted assessment measurements for blood pressure. If you have access to these blood pressure monitors, some basic instructions include:

- The blood pressure cuff is placed snugly around the upper arm, at the same height as the heart while the person is seated with their arm supported. Make sure that the correct size of cuff is used or the reading will not be accurate.
- For automatic cuffs, press the button and wait for the cuff to inflate, deflate and then report back the readings.
- Take the blood pressure in the other arm in the exact same manner. If there is a difference of more than 10 mm Hg between the measurements of the two arms, note this on the assessment card for their doctors to be aware of. This difference can mean an increased risk for cardiovascular disease.

There are several easy steps to taking blood pressure readings. A person being assessed should:

- Avoid eating, smoking and exercising for at least 30 minutes before being measured.
- If the blood pressure reading is high on the first try, wait 5 minutes and try again.

If your team does not have access to blood pressure monitors and a community member's self-assessment indicates that they are at medium to high risk levels, they should be referred for a visit to the clinic to measure their blood pressure.

Assisted assessment score

A simple scoring chart is provided on the card to allow the person to see their risk level. This assessment allows a person to see how their behaviours relate to their NCD risk level and helps their volunteer to better understand their needs. Volunteers can learn to use and monitor the assisted assessment score cards with ease. Community members with medium or high risk level results should be referred to the appropriate community resources for the clinical assessment.

Community members should have their blood pressure and BMI measured at least once a year.²⁰ If a person has a high blood pressure reading, the volunteer should measure blood pressure again to ensure an accurate reading. When conducting a home visit, offer to do an assisted assessment for all family members present. Guide each family member through completion of the self-assessment card.

Children as young as 3 years of age and up and all adults should get their blood pressure checked regularly. The safe limits for blood pressure vary by country; a general guideline is for your blood pressure measurement to be between 90/60 mmHg and 120/80 mmHg. This range means that your heart and brain is able to get good blood flow through your arteries.

¹⁹ WHO, *Executive summary – Child growth standards*.

²⁰ WHO, *Avoiding Heart Attack and Strokes*, 2005.

Clinical assessment

Clinical assessment should be done at a properly equipped clinic, laboratory, hospital or doctor's office. The assessment requires properly trained staff and sterile equipment to take blood samples for analysis. Blood samples should only be taken when a person has not eaten any food for the last 12 hours and no alcohol for 24 hours so that the test is the most accurate. The blood samples will be analysed to look for two things:

Blood sugar levels

This is the amount of sugar or glucose present in the blood and measures for diabetes. There is some glucose usually present in the blood and a small amount is acceptable.

Cholesterol

Too much cholesterol in the blood can block the flow of blood from the heart to the organs and back to the heart. Too much cholesterol could be a sign of cardiovascular disease or other NCDs.

Clinical assessment score

A simple scoring chart is provided on the card to allow the person to see their risk level based on the clinical assessment measures. This assessment allows a person to see how their behaviours relate to their NCD risk level and helps their volunteer to better understand their needs.

Volunteers should refer community members with medium or high risk level results to a health professional for further intervention.

Community members should go to a clinic to have their blood sugar tested every year.²¹

Teaching a community group to use the assessment cards

Volunteers should bring a ready supply of blank assessment cards to each community meeting and on each home visit to start work with people wanting to assess their NCD risk. To teach a group of community members how to self-assess their NCD risk, volunteers can use the following checklist:

- Show the assessment card from the toolkit for all to see and follow along.
- After conducting topics 1 and 2 with the community, they will be more aware of NCDs and the risk factors that can cause NCDs.
- Explain that knowledge is the first step to better health. In understanding their own risk level for NCDs based on their behaviours, community members can determine if they are at low, medium or high risk of getting or having an NCD and what they should do about the risks.
- Distribute assessment cards to all community members. Allow them to take extra cards, as needed. They may also choose to take some home for family members to complete.
- Show the assessment card to all.
- Explain the three parts of the card. The first section –self-assessment – they

²¹ WHO, *Avoiding Heart Attack and Strokes*, 2005.

can do today.

- Explain the questions on the self-assessment card and the answer choices in the self-assessment section.
- Model completing the self-assessment, using a marker and the risk assessment card as an example. Volunteers can use their own personal data or complete the card referring to a fictional community member. Use the NCD risk assessment card guidance sheet accompanying the cards.
- Read through the score box out loud and model how to score the card. Based on the results on the assessment card, explain that regardless of score, doctors advise that every person has their blood pressure and blood sugar checked every year. Based on this timetable, all community members are advised to go for an assisted assessment. These additional measures help to determine NCD risk more accurately.
- Ask them to privately complete their self-assessment now.
- Ask if anyone would like to share. If so, discuss their scores and next steps.

Referral for further assessment and care

- Explain the assisted assessment process and announce the time and place when volunteers are available to conduct the assisted assessments. The community volunteers should host assisted assessment sessions regularly – no less than every 2 months, so that community members can check in on a regular basis.
- Explain the clinical assessment process and provide details of clinics, doctors, hospitals and laboratories in the vicinity that can conduct the clinical assessments.
- Explain to the community that they should always consult a volunteer if they have any health concerns that require immediate attention.

Home-based psychosocial support for NCD diagnosis

Tool 3.4 - Home-based psychosocial support for NCD diagnosis

- In the event that a community member is diagnosed with an NCD, it is advised that the volunteer conducts no fewer than three home visits in the month following diagnosis to provide psychosocial support as needed, to answer any questions and to follow-up with any information.
- On home visits, the volunteer should identify the community member's primary risk factors and support them in behaviour changes to prevent further complications. Regular self-assessments and assisted assessments will help the community member to gain control of their diagnosis and work towards a healthier lifestyle.

Checking for understanding

Indicate true or false for each statement.

- The self-assessments of community members are completed by the trained volunteer. True/False
- The assisted assessment can be completed by a trained volunteer. True/False
- The clinical assessment can only be conducted once. True/False
- All three assessment sections must be completed. True/False
- Once the assessment card is completed, the trained volunteer should follow up with those community members who are at a high risk, if appropriate. True/False

Topic 4

Identification of risky health behaviours and adoption of healthier behaviours

Learning objectives

At the completion of this topic, you will be able to:

- Identify the volunteer's role as a change agent
- Discuss the stages of behaviour change
- Discuss the fluctuations in how people change their behaviours

Main learning points

- To lower risk of NCDs, behaviour change is always recommended.
- There are many interventions that can be successful in helping people to adopt healthier behaviours. These interventions are different for different contexts and determinants.
- There are five general stages in behaviour change. The steps are: knowledge, approval, intention, practice and advocacy.
- People move up and down between these stages frequently for any number of personal reasons.
- The three elements to consider when changing behaviour understands a person's motivation to change, the clarity of their plan in reaching their goal and ensuring that the person's environment supports the change.
- Behaviour change is not an exact science.
- Behaviour change requires a daily commitment to healthy behaviours.
- In every person, behaviour patterns are triggered or prompted by specific people, events or activities. Understanding these triggers helps people to be prepared to practice healthy behaviours when the triggers happen.
- Effective behaviour change messages should meet three simple criteria: Make the message simple; communicate a benefit and call to action.
- To reach the advocacy stage, a person generally needs to practice a behaviour consistently for at least two months.

Topic summary

Introduction - Stages of behaviour change poster

There are many interventions that can be successful in helping people to adopt healthier behaviours. These interventions are different for different contexts and determinants.

The IFRC Behaviour Change framework provides a rich process for determining the causes for people's unhealthy behaviours, identifying ideas of how to change the behaviours based on these causes and specific techniques. This topic is meant to provide general information about how people try to change their unhealthy behaviours and work to adopt healthier behaviours. For more specific information and guidance on writing specific behaviour change plans for community members and community groups, refer to the IFRC behaviour change framework.

Behaviour change

Tool 4.1 - Five stages of behaviour change poster



In some instances, adopting healthy behaviour may mean stopping a current habit or behaviour like NOT smoking, NOT eating unhealthy foods, NOT sitting all day on the couch or NOT drinking too much alcohol. In other instances, adopting healthy behaviour may mean starting or increasing a new habit or behaviour like eating MORE fruits and vegetables or being MORE physically active.

These two actions are very different, but a person undergoing any type of behaviour change will progress through the five recognized stages of behaviour change – stages that every person who wishes to change their behaviour will go through if they succeed.

Often, it takes a person several attempts until they are successful. They may progress up a step and then fall back. This is normal. The important thing is to pick up from where you are and continue to move forward.

Ben E. Ficiary attends an NCD session and learns so much about what an NCD is and what could happen if he stays on the couch every day after school. He never really understood why people said that he should get up and get his body active and he never understood how exactly it could hurt him. Now he understands how inactivity deprives his body of movement and activity — things his body needs to stay healthy and to fight off sickness.

Ben is at the knowledge stage.

Knowledge

Knowledge is the first stage and it is basically information about why behaviour should be changed. It can include data and statistics about the consequences associated with continuing an existing behaviour.

Ben agrees that all that sitting on the couch and watching TV is very bad for his health. **Ben is at the approval stage.**

Approval

Approval is the second stage and means that the person has heard the new knowledge and agrees that it is a good idea, that it makes sense to them. This stage does not mean that the person plans to do it personally, only that they agree that they understand the knowledge shared with them.

Note that knowledge is rarely enough to get people to change. **Knowledge is important and the first step, but it is not enough to actually get people to change.** This is where they get the information about what changes are important for a healthier life – one of the conditions for making behaviour changes. Most community sessions are only knowledge-based and so do not push people to the next levels.

Ben does not want to get sick. His father was very sick with diabetes and lost his eyesight to the disease. After taking the self-assessment, Ben sees that he is at a high risk for an NCD. Ben is scared and he has developed a roll of fat around his belly. He decides he will start walking to school every day. He will start next Monday. He makes a goal – to walk to and from school every weekday. And if it is raining, he will take an umbrella. He makes a goal – to only watch TV once he has gotten his 30 minutes of walking. He comes up with a plan for how to do it – he will pinch his belly fat every time he really does not want to go for his walk and try to remind himself that walking is good for his health. His motivation is to live a long, healthy life and to not die young like his father did. For support, his sister agrees to support him and a volunteer agrees to check in on him each week to see how he is doing, to encourage his efforts and help him stay on track. **Ben is at the intention stage.**

Intention

Intention is the third stage. Without intention, behaviour change is not possible. This stage is the first stage in behaviour change when a person takes the knowledge and personally decides that they want to apply it to their lives and that they will try to change their behaviour.

This is a powerful step as this is the stage where they identify their goal and the motivation to adopt a healthier behaviour. Motivation is one of the most powerful elements for making behaviour changes. It always helps to know what each person's motivation is when leading a group towards healthy behavioural changes as people will need to be reminded of their motivations often in the next stage. The third condition for successfully making behaviour changes is to have a supportive environment that will help people stay on track and not fail.

On Monday morning, Ben wakes up and remembers that this is the day he is walking to school. He immediately wants to sleep a little longer and take the bus, but he sticks to his goal and his plan. He quickly gets dressed and goes to school. His feet hurt but every time he wants to stop, he pinches his belly fat. Each time he wants to stop and get on the bus, he thinks of his father. He realizes how hard this will be, but he is reminded of his motivation – to live a long life and not die at a young age like his father did.

He remembers what the volunteer said about his muscles and his body and how they are already starting to get stronger after just one day of activity. When he gets home, his sister makes his favourite dinner and tells him that she is proud of him. The volunteer comes by to check on him, reminds him of how his body is healing and encourage him on his good work towards living a long, healthy life. Ben feels discouraged at how hard this is, but is proud that he made it through the day. **Ben is at the practice stage.**

Practice

Practice is the fourth stage and involves the person actually applying the behaviour. They actually DO the behaviours that they committed themselves to doing. This is the most important step. The four conditions for a person to succeed in changing a personal behaviour include:

- Defining a clear and simple goal
- Gathering sufficient and accurate knowledge to make a clear plan
- Having lots of personal motivation to follow through
- Having and relying on a supportive environment

As a volunteer, your role in helping community members is to help people stay in this stage and not fail. To support their hard work and goals, follow these guidelines with community members:

- Remind them often of their goal
- Remind them often of their motivation for reaching the goal
- Give them clear guidance on how to reach their goal
- Give them frequent reminders of how to successfully respond to challenges or triggers in their environment
- Provide support from others going through similar changes in behaviour, such as in a community group

The practice stage is the stage at which people can either fail or succeed. However, when someone fails in this stage, they will likely fall back to either the approval stage or the intention stage and with proper support can be helped to once again achieve the practice stage. The five guidelines for volunteers listed above will help you keep the majority of community members on track towards healthier behaviours.

The key to helping make behaviour change success is frequent check-ins with a support network such as the community health meetings. Regular meetings – no less than once a week is advised for the first 6 weeks of initiating a change in behaviour – are key to allowing people to have a place to share their successes, their failures and support each other as they learn a new behaviour. Support groups help people to realize that they are part of a group all struggling with the same issues and that the group often has answers to questions they may have like how to respond when someone offers you a cigarette at work after lunch or how to say no to your aunt's offer for a second and third piece of cake. Support from others going through the same struggles is essential in the first critical month when new behaviours are beginning to form.

Finally, remember that practice stage is just that – practice. NO one is perfect. All we can expect is continued practice of healthy behaviours. Soon, they will become a habit.

Ben has been active for 3 years. The first two months of getting active was the hardest and everyday he was exhausted. But he made it through. His roll of fat went away within 2 months and he now walks everywhere. **He lost weight and feels so much better about himself.** He is sleeping better and spending time with his friends doing activities outside like canoeing and hiking. Ben now provides support to others who are trying to get active. - He helps them make a clear goal, make a plan for what to do when they just want to sleep in or lay on the couch, say what their personal motivation is for making this difficult change and how he will support them as they go through the difficult first few months. **Ben is at the advocacy stage.**

Advocacy

Advocacy is the fifth stage and is home to people who have been practicing healthy behaviours consistently for at least two months to reach their goal and who share their success with others trying to reach the same goal. People in the advocacy stage guide people in the other four stages to success with motivation, inspiration and support. Red Cross Red Crescent volunteers can be seen as advocates. If the volunteer has not reached the advocacy stage personally for a behaviour change, they are still able to lead a community group but will take more of a facilitator role rather than a teacher role in leading sessions.

When a person has been successful in their practice of a new behaviour after 3 months, they can be at the advocacy stage. This does not mean that everyone is at the advocacy stage after 3 months but that this is the minimum amount of time that a person needs to successfully practice their new behaviours before they might be advocates. Each person and situation is unique.

Remind the group that while they may think they want a cigarette, a candy bar, a third beer or to sit on the couch all day, their physical body actually wants them to not smoke, to eat fruits and vegetables, to limit their drinking and to get up and get moving. Our bodies respond immediately to healthy changes!

Slides and ladders of behaviour change

Tool 4.2 and 4.3 - Slides and ladders of behaviour change kit

Behaviour change is very difficult. Even when people want to only do their best, they may struggle to maintain changes at first. Introduce the concept of moving through the stages by showing how behaviour change can take time and patience by showing the slides and ladders demonstration kit.

- Show the slides and ladders game board.
- Read aloud the scenario written on the back of the card as you move the figure up and down through the stages.

Behaviour change stages

Tool 4.4 - Behaviour change stages game kit

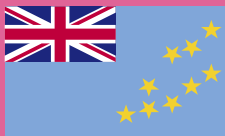
This game is appropriate for play with groups of 4 to 5 people per group – up to 30 people total. Each group gets a behaviour change stages board and a set of behaviour change cards. Groups must correctly identify the behaviour change stage of the person who made the statement on the card, placing each card at the correct stage.

Lead the full group into a discussion about why each card is placed where it is. If you have a higher literacy group, you may also distribute game packets of the behaviour stages game to teach the behaviour change stages in a fun and interactive way.

Checking your understanding

- How is the Red Cross Red Crescent unique in changing the lives of community members?
- What are the five stages of behaviour change?
- Why do people often move up and down through the stages of behaviour change?
- What are the four conditions for a person to succeed in making a behaviour change?
- What are the five ways that a volunteer can support a person making a behaviour change?

Case study:



On the tiny island of Funafuti, Tuvalu Red Cross Society has been working successfully in the Te Kavatoetoe settlement – population 500 residents. Funafuti Island is a small atoll, only 1 metre above sea level faced with overcrowding, limited land, poor soil quality all that makes it difficult to grow fruits and vegetables. All food must be imported to this tiny island and is often too expensive for many households. The limitation of land makes it difficult for people to get proper exercise – limited to playing on the local airfield in the mornings and evenings when not in use. Additionally the effects of climate change make it difficult to grow any foods– if they own land, which most people do not.

The Tuvalu Red Cross Society held a participatory community assessment that identified NCDs as a health priority. To address the many challenges faced by the people, Tuvalu Red Cross Society organized and implemented activities that addressed NCD risk factors. Activities included setting up a weekly walking group, holding sporting events for youth and organizing home gardening workshops delivered by the Ministry of Agriculture. A home gardening competition with 13 households was held with garden fencing provided by Tuvalu Red Cross Society. Vegetables and fruits are growing nicely.

There has been a sense of empowerment among those who are involved in the project. A community member who has personally benefitted and gone on to inspire the community said, “I participated in the healthy programme including fitness activities and feel much better than before. I used to weigh 130 kilos but the last time I was weighed, I was only 84 kilos.” Another young community volunteer is now helping run the Tuvalu Red Cross Society gym and is involved in all Red Cross activities. “The project inspired me to join the Red Cross as a volunteer to help vulnerable people especially in my community. Now I have a good knowledge of the programmes running and I am always ready to take part in activities, whenever the Red Cross needs my help.”

Topic 5

Development of an health action plan

Learning objectives

At the completion of this topic, you will be able to:

- State the four essential elements of effective behaviour change
- Use assessment card results to determine behaviour change goals and health action plans
- Write a personal health action plan
- Discuss the positive effects of changing risk factor behaviours
- Discuss how to best maintain healthy behaviour

Main learning points

- Applying the results of an NCD risk assessment to writing a health action plan will help you to lead a healthier life.
- The four essential elements of effective behaviour change are:
 - Defining a clear and simple goal.
 - Gathering sufficient and accurate knowledge to make a clear plan.
 - Having lots of personal motivation to follow through.
 - Having and relying on a supportive environment.
- Making behavioural changes around the four risk factors for NCDs provides immediate benefits.
- By understanding the health benefits of stopping tobacco use, limiting alcohol, eating five servings of fruits and vegetables a day and increasing physical activity to 30 minutes per day, individuals can set health goals and decide upon a behaviour change plan to respond to their assessment results.
- Health behaviour support meetings can be a powerful resource for helping people to feel supported to make necessary behavioural health changes.
- If you have already been diagnosed with an NCD, follow your doctor's advice about follow-up exams and medication schedule. Adopt healthy behaviours to avoid being diagnosed with an additional NCD.

Topic summary

Introduction

This topic involves bringing together all the information and skills from topics 1 to 4.

The essential elements of behaviour change are important pieces of coming up with a health action plan. The person wanting to make the behaviour change must:

- **Define a clear and simple goal**
- **Gather sufficient and accurate knowledge to make a clear plan**
- **Have lots of personal motivation to follow through**
- **Have and rely on a supportive environment**

These four things are all very personal and unique to each person.

The final element that ensures that people making behaviour changes stick with a new behaviour and not go back to old habits is having a supportive environment. Support in any form – whether from and with neighbours, other family members, school-based or other community-based organizations is critical to success.

A supportive environment, like a community health group, can meet regularly and provide a place for people to discuss their new changes and the challenges and successes they are experiencing is critical to ensuring that people make and stick to changes of their behaviours for the better.

In light of this, volunteers leading community groups in adopting and keeping up healthy behaviours is one of the most powerful elements in halting the epidemic of NCDs globally. The research shows that schools and community groups are effective venues for behaviour change to happen.²²

Health action plans

Tool 5.1 - Guidance for health action planning

Refresh the group about the use of the risk assessment cards using the assessment card from the toolkit.

Demonstrate how to complete your own health action plan based on your assessment card results.

There are four critical elements if you want to successfully change your behaviour:

- **Define a clear and simple goal**
- **Gather sufficient and accurate knowledge to make a clear plan**
- **Have lots of personal motivation to follow through**
- **Have and rely on a supportive environment**

These things are all very personal and unique to each person so everyone's action plan will be different.

²² Jepson, et al., *BMC Public Health*, 2010.

The first question asks for what behaviour you want to start OR stop doing. You might want to start eating more fruits and vegetables or you might want to stop smoking. You would write in this top box whatever behaviour you want to have.

Using a marker and the NCD risk self-assessment card from topic 3, write your behaviour goal directly on the card. You can use pictures or text.

You would need to think then about the reason that you really want to achieve this goal. Do you want this goal because your volunteer said you should or maybe because you want to live a long life for your children? The motivation is the reason in your heart as to why you want to achieve the goal. It is usually always personal and is rarely because someone told you to do so. The motivation usually always stays the same.

Using a marker and the NCD risk self-assessment card from topic 3, write your motivation directly on the card in the second box. You can use pictures or text.

You now need to come up with a plan. The plan lets you define how you plan to stop unhealthy behaviours and start healthy behaviours. The plan is how you plan to keep up the healthy behaviours even when you do not want to, when it is hard to keep going. The plan defines what you will do when you really want to stop because it is difficult and it will be difficult at times.

Using a marker and the NCD risk self-assessment card from topic 3, write a little about your plan. Write your plan to keep doing the healthy behaviour, what you will do when you want to stop and how you will refer to your motivation to keep going. Write also what you will do when you “slide” back – like in the slides and ladders game – and stop doing the healthy behaviour. Discuss how you will get back on track and write in the box. You can use pictures or text.

The final critical piece in planning a health action plan is to identify your support network – which you will rely on when you need support or encouragement. You can also define how and when these people will help you. You can have one friend whom you go for a walk with every day to keep you moving. You might have a friend you call every time you want to eat some unhealthy food and you really, really want to eat it but know you should not. You might go to community meetings or even visit an online support group to meet other people also struggling with your specific behaviour change.

Using a marker and a NCD self-assessment card, write a little about your support network. You can use pictures or text.

Writing a health action plan

- NCD Risk self-assessment card (completed in topic 3)
- Guidance for health action planning
- Marker
- Community members personal assessment card

Note - A health action plan details how, why and when behaviours will happen.

- **Identify a goal:** Have each community member identify a goal based on their assessment results and write the clear, specific goal at the bottom of their own assessment card. These goals can be short-term or long-term with the understanding that short-term goals are meant to build towards success of a long-term goal – adopting and maintaining healthy behaviours.

Possible goals:

- **To stop smoking**
- **To have no more than three alcoholic drinks a week**
- **To eat five servings of fruits and vegetables a day**
- **To go for a 15 minute walk three times a day**

Whatever goal they write, it should align with the assessment results on their card.

For example, if their assessment card shows that they are at high risk for NCDs and they smoke heavily or live with a smoker or in a house with a cook stove their health goal should be “to stop smoking” or “to only cook with a clean burning cook stove ” or “to ask smokers to not be in the same room as me and my children.”

They may have additional goals like “lose weight” but the assessment area showing their greatest health risk should be their primary goal. Each person may have many goals, but they should focus on only one goal at a time to make sure the new behaviour is able to sustain and become a habit.

- **Motivation:** Once they have written their behaviour change health goal, ask them to come up with their motivation or the very personal reason why they want to achieve this goal.

The motivation is something that is usually deeply personal to each person. Motivations could include wanting to live to see your children grow up, wanting to not die a similar painful death as a lost loved one who suffered from an NCD, wanting to stay attractive, wanting to lose weight or wanting to live a long and healthy life.

- **Create a plan:** The plan addresses how a person plans to change their behaviour – the routine, thoughts and support they plan to follow each day to make the change. The behaviour change will be difficult and they will often want to quit and go back to their old unhealthy behaviours. But having a plan – which includes what they will do when change proves to be very difficult – is another critical piece in helping to sustain behavioural change. The plan should include:

- a. A new routine will replace the old. This can mean that instead of a cigarette upon waking, the person does deep breathing or takes a walk. This is a healthy substitute to fill the gap left by abandoning the unhealthy behaviour.
- b. An activity that they can do anytime, anywhere to handle the stress that they may experience in making the healthy choice each time. This means that if faced with their old habit – a candy bar, a third glass of beer, an afternoon of idly sitting on the couch, or a cigarette – they instead have a quick “escape” behaviour that removes them from the

stimulus.

c. The plan needs to be easy to do and something that feels comfortable to the change-maker.

d. The plan needs to identify the change-maker's support person or support network. Regular check-ins with a support person or people provides the change-maker with a goal to work towards each day or each week, that allows them to share and get support as needed.

- **Support** – support from a person, a group of people, or even a special thing that will always remind them of their motivation.

They must next identify what their support system will be and who will be in their support system. This support needs to be in place and the people in the support network need to be aware of their responsibility to support and encourage when times get tough.

These four pieces make up the health change action plan. It is not enough to say: I will eat five servings of vegetables and fruits every day. A health action plan details how, why and when that will happen.

Previous NCD diagnosis

If a person has already been diagnosed with an NCD, they should:

- **Follow their doctor's advice about follow-up exams and medication schedules. Cost-effective treatments are available.**
- **Adopt healthy behaviours to avoid complications or diagnosis of a secondary NCD.**²³

Using a marker and the assessment card, the Red Cross Red Crescent volunteer would model the writing of the four boxes at the bottom of the card in the health change action plan.

To model the writing of the plan, pre-complete the assessment card with your scores from topic 3.

Identify a goal based on the results used to model the completion of the self-assessment (from topic 3) and give as much detail as possible as you write the components of the plan.

You may choose to use one or more of the following examples to help you to model the writing of your plan. Each example highlights the scores every person received on their self-assessment in the left column, the person's problem behaviour where they are focusing their change in the middle column and an example of what a corresponding health action plan looks like in the right column.

²³ WHO, *Protocols for health promotion, prevention and management of NCDs at primary care level, 2013.*

	Manee's self-assessment scores:	Manee says:	Manee's health action plan:
Healthy diet	①	I want to stop feeling tired all the time and I want to lose this weight. I always feel poorly and I want to feel good again. I have a small child, Rafael, who I cannot keep up with and I am sad when I cannot play with him. My plan is to eat one serving of a vegetable at every meal and have a piece of fruit after lunch and dinner. I will make sure to buy enough fruit at the market each week to have plenty on hand. I will attend the Red Cross Red Crescent meeting each week to share recipes, see how others are doing; get inspiration when it feels like it's just not worth it any longer and to feel part of the group.	Goal: Eat at least five fruits and vegetables a day. Motivation: I want to be able to play with my young child. Plan: Buy plenty of fruit and vegetables several times a week; get rid of the bad foods in my kitchen. When I want something unhealthy, I will do a yoga stretch and remind myself that I want to play with Rafael. Support: Red Cross Red Crescent group meeting every Monday and Thursday, daily talk with my sister who is also trying to eat healthy, support from my husband.
Physical activity	②		
Tobacco use	②		
Excess alcohol	②		
Sex and age	YES		
Family history	YES		

	Sanjay's self-assessment scores:	Sanjay says:	Sanjay's health action plan:
Healthy diet	②	I want to feel better – I wake up every morning with a full-on headache and feel ill all morning every day. I am too young to feel this old. My mother died from drinking too much and I do not want to die like that. My plan is to only drink half a glass of beer each night. When I finish, I will immediately wash out my glass to signal that I am finished. If my friends push me to have more, I will tell them I am not thirsty. If they continue to push me to drink, I will have to limit my time out with them. It is not worth my life to give in to them. I will attend the Red Cross Red Crescent meeting every week to get support (I will need it!), to talk with others about how to stop feeling pressured to drink all the time and be reminded why I am drinking less, maybe even meet some new friends.	Goal: Drink not more than half a beer each day. Motivation: I do not want to die like my mother. I want to feel better. I want to feel young. Plan: Wash out my glass after I drink my half of a glass and put the glass away. I will put off peer pressure by telling my friends no or limit time spent with them. Support: Weekly group meeting, daily call with my brother who gave up drinking last year.
Physical activity	②		
Tobacco use	②		
Excess alcohol	①		
Sex and age	YES		
Family history	NO		

	Anna's self-assessment scores:	Anna says:	Anna's action plan:
Healthy diet	3	I want to stop coughing every morning. It even wakes me up every night. It hurts and I am afraid of what my lungs look like after all those cigarettes.	Goal: Stop smoking cigarettes completely.
Physical activity	2		Motivation: My heavy cough is painful. I am afraid of the damage I am doing to my baby. I want to feel and look good like my brother, Kamil who gave up cigarettes.
Tobacco use	-1	My baby has just started to have a bad cough now every time I smoke – my husband too. I am worried that my smoking is hurting them and making my son cough. I think this is why I need to stop – I cannot hurt them. And if I go outside, I cannot play with him. Smoking is hurting me and my family. My plan is to stop smoking cigarettes completely. Every time I want to smoke, I will talk to my brother Kamil who will help me get through. He gave up cigarettes 2 years ago and he looks so much better and healthier. He says he feels better and I want to feel better too. He said he would help me get through the rough times and maybe I will join the meeting if the times work out for me.	Plan: I will throw away all my cigarettes. Every time I want a cigarette, I will put the money I would have spent to buy a cigarette (50 cents) plus \$1 that I would have had to spend on a doctor bill for my cough if I had smoked it – into a jar on my table. Whenever someone offers me a cigarette, I will think of that jar of money and how I plan to spend it, taking my family on a small holiday next summer. We never did that before, because we have no money. But I can do this. If I really, really want to smoke, instead I will go for a walk to Kamil's house.
Excess alcohol	2		
Sex and age	YES		
Family history	YES		Support: Talk to Kamil who said he would support me. Maybe attend a group meeting. Play with my baby instead of smoking.

	Yohannes' self-assessment scores:	Yohannes says:	Yohannes' action plan:
Healthy diet	1	I want to stop feeling badly. I haven't done much exercise because I am heavy and I am embarrassed. I am at a high risk for diabetes and I do not want to lose my sight, have to take insulin and become sick. I have two children who I want to see grow up. My plan is to walk to the market every day for fresh fruit. If it is raining, I will do simple exercises in my home. I will attend the meeting because it is near the market and the Red Cross Red Crescent lady said the meetings would be for people with the same health goal as me.	Goal: Walk every day. Eat fruits I buy at the market.
Physical activity	1		
Tobacco use	3		Motivation: I want to feel better. I do not want to lose my sight, have to take insulin and become sick
Excess alcohol	3		
Sex and age	YES		Plan: Walk to the market every day for fresh fruit. If it is raining, I will do simple exercises in my home.
Family history	YES		Support: Attend a Red Cross Red Crescent group meeting.

Healing your body through behaviour change

Tools 5.2, 5.3, 5.4 and 5.5 - 'Effects of...' posters

Effects of changing behaviours of diet, alcohol, activity and tobacco are shown in four posters:

- Effects of quitting tobacco
- Effects of drinking less alcohol
- Effects of a healthy diet
- Effects of physical activity

In topics 1 and 2, we discussed the dangers of NCDs and how the Dangerous Four risk factors damage and kill. After you stop using tobacco and drinking too much alcohol and when you start eating a healthy diet and getting physical activity, there are immediate positive changes in your body. While you may think you want a cigarette, a candy bar, another beer or to sit on the couch all day, your physical body actually wants you to not smoke, to eat fruits and vegetables, to limit your drinking and to get up and get moving. Your body responds immediately to healthy changes.

The four posters in the topic 5 toolkit show and discuss the effects of physical activity, of quitting tobacco, of drinking less alcohol and of a healthy diet. The posters feature how the body improves once you begin to stop engaging in unhealthy behaviours.

Follow the guidance on the backs of these posters to facilitate discussion on each of the four risk factors.

Community-based support

The assessment card helps you to teach community members to write individual plans of action. To make changes at the community level, help community members to work together to come up with plans of action that involve other community members and start to weave support networks between the community.

Stopping a behaviour that has been practiced for a long time will be difficult at first but by working together with a community group such as through Red Cross Red Crescent, people have the support and resources they need to make the changes for a healthier life. Starting a new behaviour may seem a bit easier initially than stopping an existing behaviour, but people usually have a difficult time maintaining the new behaviour. Working together with a community group has been shown to give people the necessary support and resources to continue the healthy behaviours when they most feel like giving up and going back to their old unhealthy – and possibly deadly – behaviours.

Preventing NCDs in the long run

When working with community groups on NCD prevention, remember that the work is long-term. There are no quick methods to changing behaviours or to preventing noncommunicable diseases. Prevention is a lifelong process.

There are many types of support available that also involve healthy activities. These might include a football club, a cooking club, a gardening group, a walking/running club, a bicycling group, etc. There are also groups for people with NCDs needing support and guidance such as a cancer care group, a diabetic group, quitter's club for quitting tobacco or an alcohol recovery programme, etc. There are even online support groups available to those with the necessary resources. These groups provide a sense of community as they engage people in healthy behaviours and encourage members to share how they stay on the path to positive behaviour change.

When planning a group or activity focused on starting and/or maintaining positive behaviour change in your community:

1. Only focus on one healthy behaviour per group session. Have meetings titled with the behaviour that is being focused on and hold them at regular days and times so people remember.
2. Do not try to have community members change all of their behaviours at one time – it is not effective. Trying to change a single behaviour is difficult enough – by trying to change too many at once almost always makes the person revert back to their old familiar behaviours out of frustration. However, note that behaviours are often interlinked (e.g. smoking and alcohol use; physical activity and healthy eating) and changing one behaviour may impact on one or more related behaviours.²⁴
3. If there is some disagreement of the priority order for tackling behaviour change of risk factors, it is recommended to first tackle physical activity – a behaviour that may be easier and more attainable for most community members.
4. It is critical that you maintain the meetings on the schedule shared with the community members. Do not allow other events to disrupt the schedule. If the

²⁴ Jepson, et al., *BMC Public Health*, 2010.

support group is not reliable, community members who are in the process of changing their behaviours will lose their motivation when their support team is not available as planned.

5. If there is a meeting with little attendance, it is suggested that home visits be conducted to find out how your members are doing and why they were unable to attend. This additional step helps the group members to feel their efforts are important and that they do have the support they need to continue with changing their behaviours.
6. If a group member reverts back to their old unhealthy behaviour, reassure them that this is okay, that the stage of behaviour change is “practice,” not “perfection.” Reassure them that they can start again right now and that the group is there to support them. Refresh them on the timeline of how their body responds to the healthy behaviour and how their unhealthy behaviour puts them at risk for an NCD.
7. For unhealthy behaviours that involve an addictive substance like nicotine in tobacco or alcohol or drug use, it is recommended that you refer community members for professional help. In some instances, volunteers and staff have been trained by professionals to help guide community members through withdrawal and on to healthy behaviours.
8. Check to see if there are existing support groups and health campaigns supported by your Ministry of Health. Link your support groups or activities with these campaigns for maximum effect and exposure.

When running a behaviour change session, use the following format:

- Check-in with group to see how people did since the last time the group met.
- Ask the group to share their successes and challenges in their behaviour change.
- Ask how many people have a behaviour change plan and if this plan helped them. If it did not, discuss changing the plan.
- Discuss motivations and how people’s motivations helped get through the challenges. If the motivations did not help, discuss how to identify their true motivation(s).
- Ask the group members to share tips for overcoming common obstacles and struggles. Every person will likely have some new tip or idea that came to them as they struggled with their behaviour change. Someone may suggest a good plan, a plan to handle difficult situations, peer pressure or a craving. Have people share to encourage people who are doing well and to give new ideas to people who are struggling with the changes.
- Conduct a quick 15 minute session to refresh or teach the group knowledge about NCDs and how the Dangerous Four behavioural risk factors contribute to their onset. Use the toolkit from topics 1 and 2 to teach the information. Even if the information has been previously taught it typically takes seven actual uses of the information in a real setting for the information to truly be learnt and understood.
- Presentation or review of changes that are happening in their body due to their new behaviours. Use the “Effects of..” tools in the toolkit to teach the group how their body is responding to their behaviour. Even if the behaviour may feel difficult, their bodies are improving every minute that they maintain their behaviour change.

- Discuss the stages of behaviour change and discuss with the group how they feel they are progressing on the changes ladder.
- Hold a group discussion to review their motivations for the change, how they are making their environment safe for their behaviour change and reviewing knowledge about the positive effects of their behaviour change. These discussions can be done in homogenous groups – housewives in one group, men in another, adolescents by gender in another; religious affiliations. Any grouping that makes people feel more comfortable in sharing their motivations and progress is good.
- Wrap-up and give details of next meeting

If there are community members that miss more than one session consecutively, a home visit is advised to follow up and bring the person back into the safety and support of the group.

After the first five group sessions, it is suggested that the volunteers encourage the group members to self-assess themselves again.

As suggested previously, the volunteer group in the community should host regular days and times per week for community members to drop in for an assisted assessment. The group members should also be given an assisted assessment every 2 months to determine progress on their BMI and/or blood pressure readings. Doctors recommend that people go for blood pressure checks every 2 years as a minimum after the age of 3²⁵, but in the beginning stages when healthy behaviour is trying to become habit, checking more often can be helpful.

If appropriate, group members may also be referred for follow-up assessments as needed, particularly if they were assessed previously to be at high risk for NCDs prior to the community sessions.

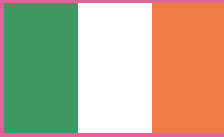
Continue with weekly meetings indefinitely to provide a safe and supportive environment to continue the positive behaviours. After several months, decide if there are any group members that have progressed to the advocacy stage and who are accepted by the group. Ask them to co-facilitate meetings, if possible.

Checking your understanding

- What are the four elements of effective behaviour change?
- How can the community volunteers support people in changing their behaviour?
- How many times a week should behaviour change support group meetings take place at a minimum?
- How many health behaviour changes should a person focus on at one time?

²⁵ NIH, *A Pocket Guide to Blood Pressure Measurement in Children*, 2007

Case study:



In June 2009, Ireland became the first country in the world to introduce a community-based health and first aid (CBHFA) approach into a prison setting at Wheatfield Prison in Dublin. The programme operates under a partnership between the Irish Red Cross, Irish Prison Services and Vocational Education Committees. Between 2010 and 2013 it was expanded to 10 prisons benefitting almost 3,500 prisoners and by 2014 will be effective in all 14 prisons nationally.

The CBHFA programme is working to tackle NCDs through peer-to-peer education provided by inmates working as Irish Red Cross volunteers. Each prison has a community health committee formed by prison staff and prisoner volunteers. The volunteers are advocates as they work to change the mindset of the community members from reactive to proactive regarding their health.

Community volunteers lead health training sessions to talk about all topics of concern such as noncommunicable diseases and the related risk factors. As a result, the communities identify health areas of focus including quitting tobacco, healthier eating and more physical activity. The activities led by the community volunteers include smoking cessation sessions to help community members quit smoking, regular times for assisted assessments to measure NCD risk including blood pressure readings and BMI measurements and physical activity sessions for inmates. The volunteers work with the community health committee for healthier options in their environment, including healthier menu options at mealtime and healthier choices of snacks and other provisions in the prison shop.

The volunteers have also led successful projects on raising awareness about personal, in-cell and prison hygiene, tuberculosis prevention, HIV and AIDS awareness and testing, winter vomiting bug, seasonal flu, dental hygiene, hepatitis vaccinations, heart disease and stroke and drug awareness campaigns. They have also initiated practical demonstrations in cardiopulmonary resuscitation and basic first aid around the prisons.

The impact of CBHFA has shown that all volunteers in the prisons have shared that they have felt personally empowered and have noticed positive changes in themselves physically and mentally. "I used to treat people the way they treated me and worked for myself independently. After volunteering, I believe in myself, I can do better. I do more volunteer work and help others – money is not important. I am less selfish. I am loyal to the team. I value life!" explains one volunteer inmate.

The fundamental principles of the international Red Cross and Red Crescent movement

Humanity The international Red Cross and Red Crescent movement, born of a desire to bring assistance without discrimination to the wounded on the battlefield, endeavours, in its international and national capacity, to prevent and alleviate human suffering wherever it may be found. Its purpose is to protect life and health and to ensure respect for the human being. It promotes mutual understanding, friendship, cooperation and lasting peace amongst all peoples.

Impartiality It makes no discrimination as to nationality, race, religious beliefs, class or political opinions. It endeavours to relieve the suffering of individuals, being guided solely by their needs, and to give priority to the most urgent cases of distress.

Neutrality In order to enjoy the confidence of all, the movement may not take sides in hostilities or engage at any time in controversies of a political, racial, religious or ideological nature.

Independence The movement is independent. The National Societies, while auxiliaries in the humanitarian services of their governments and subject to the laws of their respective countries, must always maintain their autonomy so that they may be able at all times to act in accordance with the principles of the movement.

Voluntary service It is a voluntary relief movement not prompted in any manner by desire for gain.

Unity There can be only one Red Cross or Red Crescent Society in any one country. It must be open to all. It must carry on its humanitarian work throughout its territory.

Universality The international Red Cross and Red Crescent movement, in which all societies have equal status and share equal responsibilities and duties in helping each other, is worldwide.

For more information on this IFRC publication, please contact:

**International Federation of Red Cross and
Red Crescent Societies**

P.O. Box 372

CH-1211 Geneva 19

Switzerland

Telephone: +41 22 730 4272

Telefax: +41 22 733 0395

Email: secretariat@ifrc.org



**International Federation
of Red Cross and Red Crescent Societies**

www.ifrc.org Saving lives, changing minds.